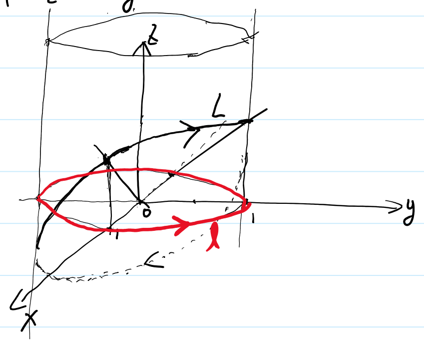
第19讲. 第Ⅱ型曲线积分(续). 格林公式

作业: P155. 1. 2.②、③. 3. 4. 5. 6. 7.

例1. 计算 $I=\int_{L^+}(y-z)dx+(z-x)dy+(x-y)dz$

$L^+:\begin{cases}x^2+y^2=1\\ z=x+y\end{cases}$ 从z轴正向向原点看去为逆时针方向.

解:



$l^+:\begin{cases}x=\cos\theta\\ y=\sin\theta\end{cases}$ $\theta: 0\to 2\pi$.

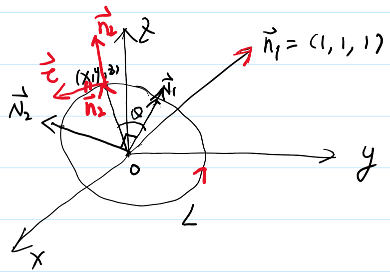
$$I=\int_{l^+}\cancel{-xdx}+ydy+(x-y)(dx+dy)=\int_{l^+}xdy-ydx$$

$$=\int_0^{2\pi}(\cos^2\theta+\sin^2\theta)d\theta=2\pi.$$

例2. 计算 $I=\int_{L^+}(y-z)dx+(z-x)dy+(x-y)dz=\int_L(y-z,z-x,x-y)\cdot\vec{\tau}\,ds$

$L^+:\begin{cases}x^2+y^2+z^2=1\\ x+y+z=0\end{cases}$ 从(1,1,1)方向向原点看去 为逆时针.

解:



$\vec{n}_2,\vec{\tau},\vec{n}_1$

$\vec{\tau}\ \vec{n}_1\ \vec{n}_2$

$\vec{n}_1,\vec{n}_2,\vec{\tau}$

$$\vec{\tau}=\frac{\vec{n}_1\times\vec{n}_2}{\|\vec{n}_1\times\vec{n}_2\|}$$

$(x,y,z)=\vec{r}$

$\vec{k}=(0,0,1)$

$\vec{N}_1=\vec{k}\times\vec{n}_1$ $\quad\vec{N}_1\perp\vec{n}_1$

$\vec{N}_2=\vec{n}_1\times\vec{N}_1$ $\quad\vec{N}_1,\vec{N}_2,\vec{n}_1$

$$\vec{r}=\frac{\vec{N}_1}{\|\vec{N}_1\|}\cdot\cos\theta+\sin\theta\frac{\vec{N}_2}{\|\vec{N}_2\|}$$

$$\vec{n}_1\times\vec{n}_2=\begin{vmatrix}i&j&k\\1&1&1\\x&y&z\end{vmatrix}$$

$$=(z-y,\ x-z,\ y-x)$$

$$\therefore I=\int_L(y-z,z-x,x-y)\cdot\frac{(z-y,x-z,y-x)}{\sqrt{(x-z)^2+(z-y)^2+(x-y)^2}}ds$$

$$=-\int_L\left[(y-z)^2+(z-x)^2+(x-y)^2\right]ds$$

$$2y^2+2x^2+2z^2-2yz-2xz-2xy$$

$$=2-\left[(x+y+z)^2-(x^2+y^2+z^2)\right]$$

$$\int_L [(y-z)+(z-x)+(x-y)]\, ds$$

$$= -\sqrt{3}\int_L ds = -2\sqrt{3}\pi$$

$$= 2 - [\underbrace{(x+y+z)^2}_{0} - \underbrace{(x^2+y^2+z^2)}_{1}]$$

$$= 3$$

例3. 求 $I = \int_{L^+} x\,dx + y\,dy + z\,dz$. $L^+: \begin{cases} x^2+y^2+z^2=a^2 \\ x+y+z=0 \end{cases}$

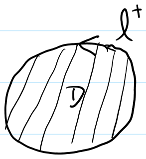
从(1,1,1)看去为逆时针.

$$= \int_{L^+} \underbrace{(x,y,z)\cdot\vec{\tau}}_{0}\, ds$$

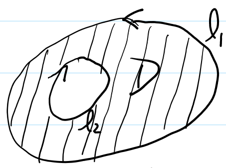
$$= 0.$$

二. 格林公式.

定义1. 设 $D \subset \mathbb{R}^2$ 区域. 其边界 $\partial D = l$ 是由n条光滑组成. 若沿着 l 行走时, D总落在 l 的左侧. 该方向定义为 l 的正方向. l^+



$\partial D = l$

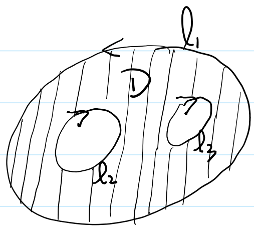


$l = \partial D = l_1 \cup l_2$.

l_1 逆时针, l_2 顺时针.

l_1^+ 逆时针. $l^+ = l_1^+ \cup l_2^-$

l_2^+ 逆时针.



$l = \partial D = l_1 \cup l_2 \cup l_3$.

定义 $\begin{cases} l_1^+: 逆时针 \\ l_2^+: 逆时针 \\ l_3^+: 逆时针. \end{cases}$ $l^+ = l_1^+ \cup l_2^- \cup l_3^-$

注: 沿封闭曲线积分. 默认为曲线正方向.

定理1. 设 $D \subset \mathbb{R}^2$ 有界闭区域. $P(x,y), Q(x,y) \in C^1(D)$. $\partial D = l$ 是光滑或分段光滑封闭曲线. 则

$$\oint_{l^+} P(x,y)\,dx + Q(x,y)\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

$$= \iint_D \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ P & Q \end{vmatrix} dx\,dy$$

$$\oint_{l^+} P\,dx + Q\,dy = \oint_{l^+} P\,dx + \oint_{l^+} Q\,dy$$

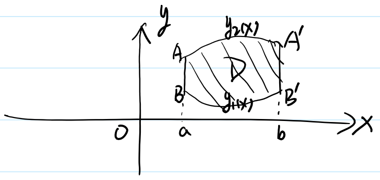
$$\begin{cases} \oint_{l^+} P\,dx = -\iint_D \frac{\partial P}{\partial y}\, dx\,dy \\ \oint_{l^+} Q\,dy = \iint_D \frac{\partial Q}{\partial x}\, dx\,dy \end{cases}$$

注: $P(x,y), Q(x,y) \in C(D)$

$$\oint_{l^+} Q\,dy = \iint_D \frac{\partial Q}{\partial x}\,dx\,dy$$

证：1°. $D = \{(x,y) \mid a \le x \le b,\ y_1(x) \le y \le y_2(x)\}$.

$y_1(x),\ y_2(x) \in C^1[a,b]$.



$l = \partial D$.

$$\oint_{l^+} P\,dx = \underbrace{\int_{\overline{AB}} P\,dx}_{=0} + \int_{\widehat{BB'}} P\,dx + \underbrace{\int_{\overline{B'A'}} P\,dx}_{=0} + \int_{\widehat{A'A}} P\,dx$$

$$= \int_a^b P(x, y_1(x))\,dx - \int_a^b P(x, y_2(x))\,dx = \int_a^b \big(P(x,y_1(x)) - P(x,y_2(x))\big)\,dx$$

$\widehat{BB'}$: $\begin{cases} x = x \\ y = y_1(x) \end{cases}$ $x: a \to b$

$\widehat{A'A}$: $\begin{cases} x = x \\ y = y_2(x) \end{cases}$ $x: b \to a$

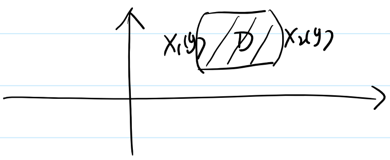
$$= \int_a^b P(x,y)\Big|_{y_2(x)}^{y_1(x)}\,dx$$

$$= \int_a^b \Big(\int_{y_2(x)}^{y_1(x)} \frac{\partial P}{\partial y}\,dy\Big)\,dx$$

$$= -\int_a^b \Big(\int_{y_1(x)}^{y_2(x)} \frac{\partial P}{\partial y}\,dy\Big)\,dx$$

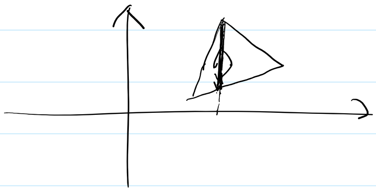
$$= -\iint_D \frac{\partial P}{\partial y}\,dx\,dy$$

2°. $D = \{(x,y) \mid c \le y \le d,\ x_1(y) \le x \le x_2(y)\}$

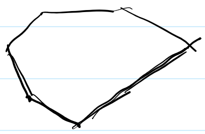


$$\oint_{l^+} Q\,dy = \iint_D \frac{\partial Q}{\partial x}\,dx\,dy$$

3°.



4°.



例1. 设 L^+ : $x^2+y^2=1$. 逆时针.

求 $\displaystyle\int_{L^+} \frac{x\,dy - y\,dx}{3x^2+7y^2}$

解:

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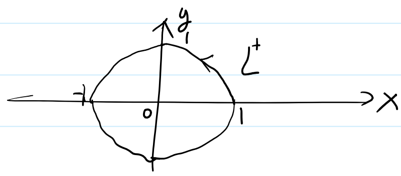

$$P(x,y) = \frac{-y}{3x^2+7y^2}. \qquad Q(x,y) = \frac{x}{3x^2+7y^2}.$$